FINAL PROJECT REPORT

India's Agricultural Crop Production Analysis (1997–2021)

1. Introduction

1.1 Project Overview

Agriculture is one of the most important sectors of the Indian economy and plays a significant role in employment, food security, and economic development. India produces a wide variety of crops across different states, seasons, and geographical regions. Understanding agricultural production patterns is important for identifying production trends, major producing states, crop-wise performance, and changes in cultivated area and yield over time.

The project **"India's Agricultural Crop Production Analysis (1997–2021)"** focuses on analyzing agricultural crop production data of India over a period of 25 years. The project uses **Tableau** to transform raw agricultural data into interactive and visually appealing dashboards and stories.

The analysis presents agricultural production from different perspectives, including year-wise production, state-wise production, crop-wise cultivated area, agricultural yield, production by unit type, leading states, and season-wise production.

The interactive dashboard allows users to explore the data using filters such as **Year, State, Season, and Crop**. These filters make it possible to analyze specific portions of the dataset and compare agricultural performance across different dimensions.

The final dashboard, titled **"Agriculture Production Insights,"** combines multiple visualizations such as maps, bar charts, line charts, pie charts, donut charts, and summary tables into a single interactive analytical interface.

1.2 Objectives

The main objectives of this project are:

1. To analyze India's agricultural crop production from 1997 to 2021.
2. To identify trends in total agricultural production over the years.
3. To compare agricultural production across different Indian states.
4. To analyze the cultivated area of different crops.
5. To understand changes in agricultural yield over time.
6. To identify the leading states in agricultural production.
7. To analyze agricultural production according to different seasons.

8. To compare production according to different production units.
9. To provide an interactive dashboard for exploring agricultural data.
10. To present meaningful insights from complex agricultural data using Tableau visualizations.
11. To support data-driven understanding of agricultural production patterns in India.

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2. Project Initialization and Planning Phase

2.1 Define Problem Statement

Project Initialization and Planning Phase

Date	22 August 2026
Team ID	xxxxxx
Project Name	India's agricultural crop production analysis (1997-2021)
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer	Select the best crop	It is difficult to predict crop production	Production varies by region, season and other factors.	Confused and uncertain
PS-2	Agricultural researcher	Analyse crop production data	Comparing large amounts of data is difficult	Data varies across crops, states and years	Frustrated and confused

2.2 Project Proposal (Proposed Solution)

Project Initialization and Planning Phase

Date	12 August 2026
Team ID	xxxxxx
Project Title	India’s agricultural crop production analysis (1997-2021)
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To Analyze India’s agricultural crop production data from 1997 to 2021. It includes data cleaning, analysis and comparison of crop production across different years, crops and regions, followed by visualization through an interactive dashboard.
Scope	The project covers agricultural crop production in India from 1997 to 2021. It includes data cleaning, analysis and comparison of crop production across different years, crops and regions, followed by visualization through an interactive dashboard.
Problem Statement	
Description	Agricultural production data is large and difficult to analyze manually, Farmers, researchers and planners may find it difficult

	to identify crop production trends and compare production across different years, crops and regions.
Impact	Analyzing the data can help users understand historical crop production trends identify high-and low- production crops or regions, and support better agricultural planning and decision-making.
Proposed Solution	
Approach	Collect and clean agricultural crop production data from 1997-2021 perform exploratory data analysis, calculate relevant statistics and compare production across years, crops and regions. The analyses data will be presented using interactive charts and dashboards in Tableau for easy interpretation.
Key Features	• Years- wise crop production analysis • Crops- wise production comparison • State/ region-wise analysis • Identification of production trends • Interactive Tableaus charts and dashboard

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	2 x NVIDIA V100 GPUs
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask

Libraries	Additional libraries	scikit-learn, pandas, numpy
Development Environment	IDE, version control	Jupyter Notebook, Git
Data		
Data	Source, size, format	Kaggle dataset, 10,000 images

2.3 Initial Project Planning

Initial Project Planning Template

Date	22 August 2026
Team ID	xxxxxx
Project Name	India's agricultural crop production analysis (1997-2021)
Maximum Marks	4 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data collection	USN-1	Downloading dataset	2	High	Rajat tiwari	12 Aug	12 Aug
Sprint-1	Data collection	USN-2	loading the dataset	1	High	Rajat tiwari	12 Aug	12 Aug
Sprint-1	Data preparation	USN-3	Clean the data	3	Low	Rajat tiwari	13 Aug	13 Aug

Sprint-1	Data visualization	USN-4	Developing visualization	5	High	Rajat tiwari	14 Aug	17 Aug
Sprint-2	Dashboard	USN-5	Creating dashboard	3	Medium	Rajat tiwari	18 Aug	18 Aug
Sprint-2	Story	USN-6	Creating story	3	medium	Rajat tiwari	19 Aug	19 Aug
Sprint-2	Documentation and recording	USN-7	Creating demo	3	High	Rajat tiwari	20 Aug	23 Aug
Sprint-2	Documentation and recording	USN -8	Creating docs	5	High	Rajat tiwari	23 Aug	27 Aug

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

Data Collection and Preprocessing Phase

Date	14 August 2026
Team ID	xxxxxx
Project Title	India’s agricultural crop production analysis (1997-2021)
Maximum Marks	2 Marks

Data Collection Plan Template

Section	Description
Project Overview	The project analyzes India agricultural crop production data from 1997-2021 to identify production trends, compare different crops and regions, and present the findings through data visualization.
Data Collection Plan	The data will be collected from published from available agricultural datasets and government/ open-data sources.
Raw Data Sources Identified	Agricultural crop production data containing inform anting such as year, state/ region, crop name and production quantity for the period 1997-2021

Raw Data Sources Template

Source Name	Description	Location/URL	Format	Size	Access Permissions
Dataset 1	India’s agricultural crop production	..\..\..\..\..\Downloads\India’s Agricultural	excel	14 mb	Public

	data form 1997-2021..	Crop Production Analysis(1997-2021).xlsx			

3.2 Data Quality Report

Data Collection and Preprocessing Phase

Date	15 August 2026
Team ID	xxxxxx
Project Title	India's agricultural crop production analysis (1997-2021)
Maximum Marks	3 Marks

Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Source	Data Quality Issue	Severity	Resolution Plan
Dataset	Missing values are present in some records/ fields.	Moderate	Identify missing values and remove incomplete records or replace values where aproriate.

Dataset	Some crop/state names may have inconsistent spelling or formatting	moderate	Standardize crop and state names to maintain consistency.

3.3 Data Exploration and Preprocessing

Data Collection and Preprocessing Phase

Date	15 August 2026
Team ID	xxxxxx
Project Title	India's agricultural crop production analysis (1997-2021)
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	**Description**
Data Overview	The dataset contains India's agricultural crop production data from 1997-2021, including information such as year, state/region, crop and production.
Data Cleaning	Checked and handled missing values, duplicate records and inconsistent crop/state names.
Data Transformation	Filtered and sorted the data, organized it by year, crop and state/ region, and prepared

	calculated fields required for analysis and visualization.
Data Type Conversion	Converted year into an appropriate numerical data format and production values into numerical format for accurate analysis.
Column Splitting and Merging	Split combined fields such as crop state information where required and merged relevant columns to Crete a structured dataset.
Data Modeling	Defined relationships between relevant fields. Tables and created calculated measures required for crop production analysis and Tableaus visualizations.
Save Processed Data	Saved the cleaned and proce3ssed dataset in csv excel format for further analysis and tableaus dashboard development.

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4. Data Visualization

4.1. Framing Business Questions

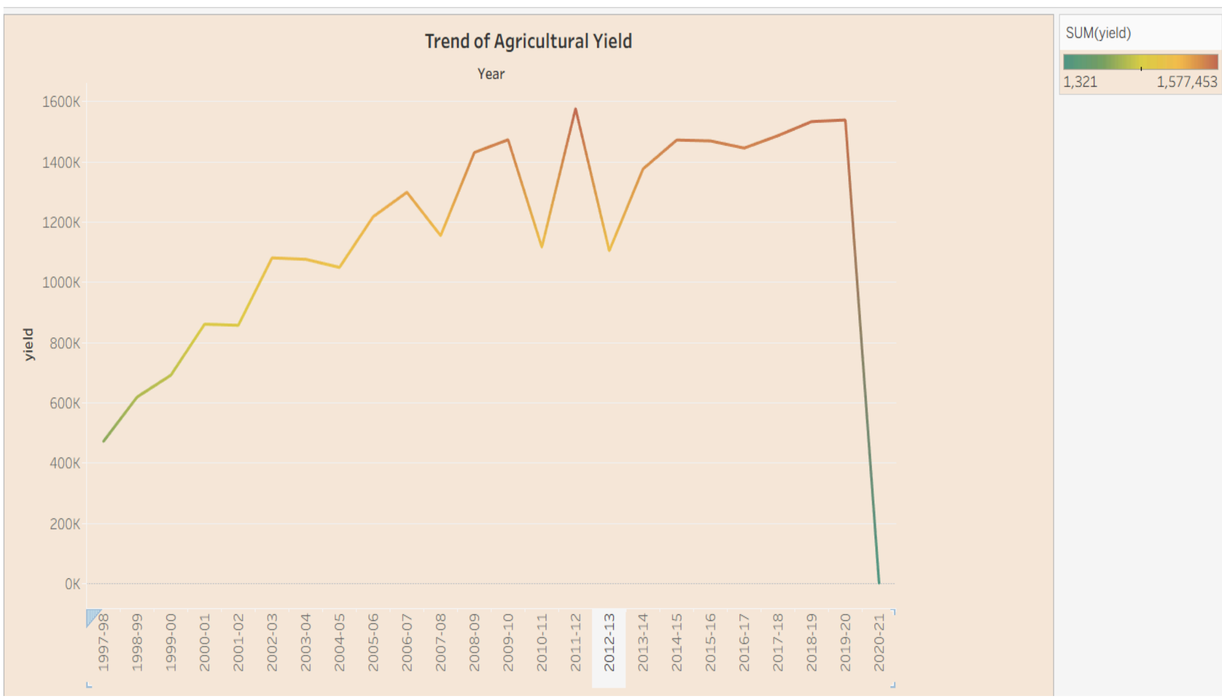
Business Question and Visualization Report

Date	18 August 2026
Team ID	xxxxxx
Project Name	India's agricultural crop production analysis (1997-2021)
Maximum Marks	5 Marks

Q.1 How has agricultural yield changed over the years from 1997 to 2021?

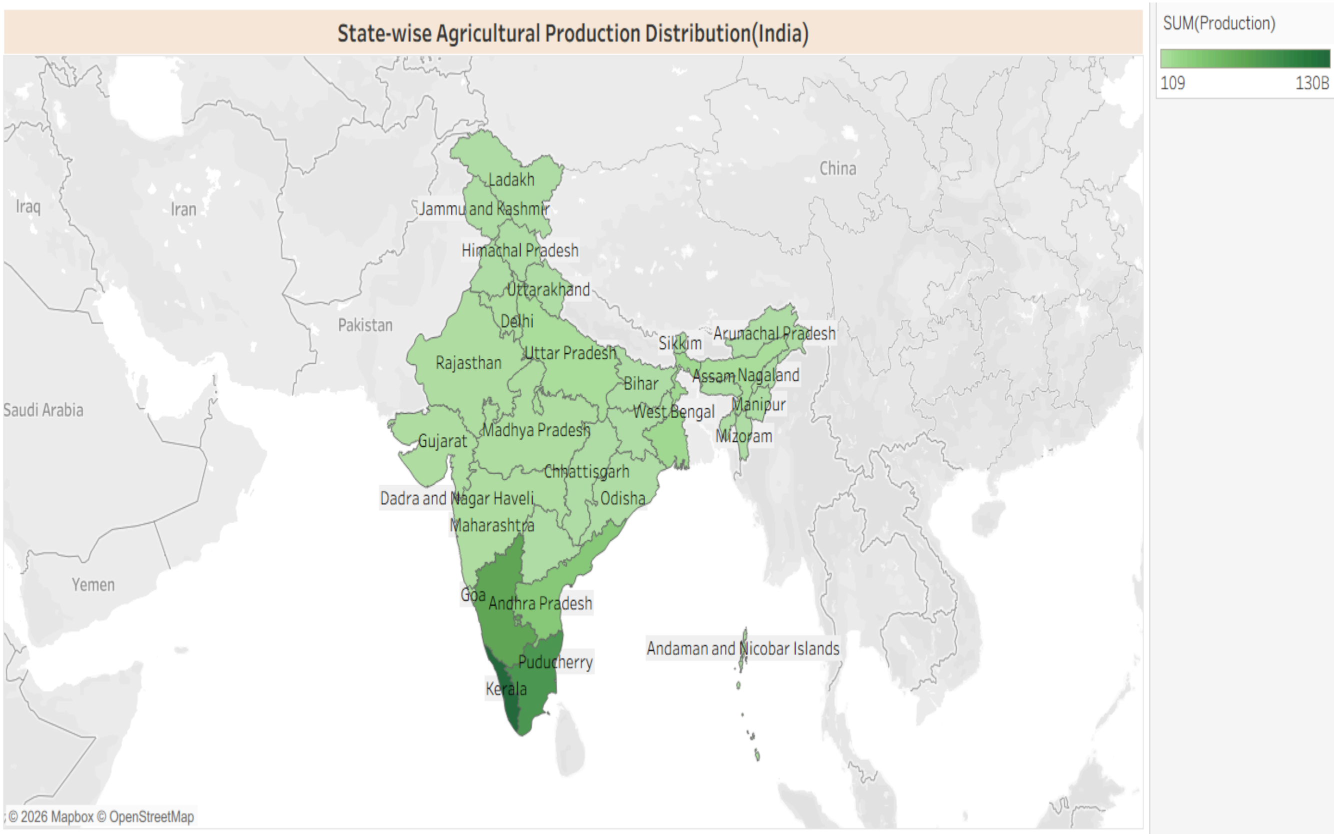
Visualizations : line chart showing the year-wise trend or total agricultural yield.

The chart helps identify years of increasing /decreasing agricultural yield and the overall production trend over the period.



Q.2 How is agricultural production distributed across different states of India?”

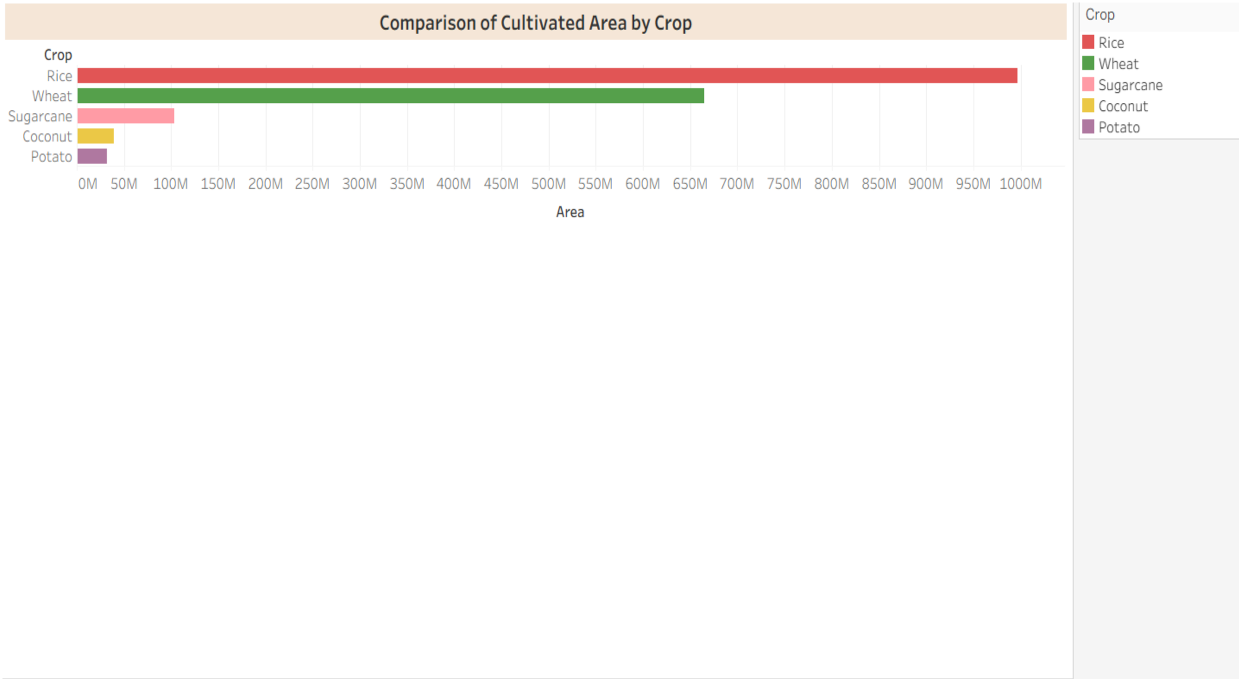
Visualization: Map showing state- wise total agricultural production in India.



Q.3 Which crops occupy the largest cultivated area in India, and how does the cultivated area vary among different crops?

Visualization- Horizontal bar chart comparing the cultivated area of different crops.

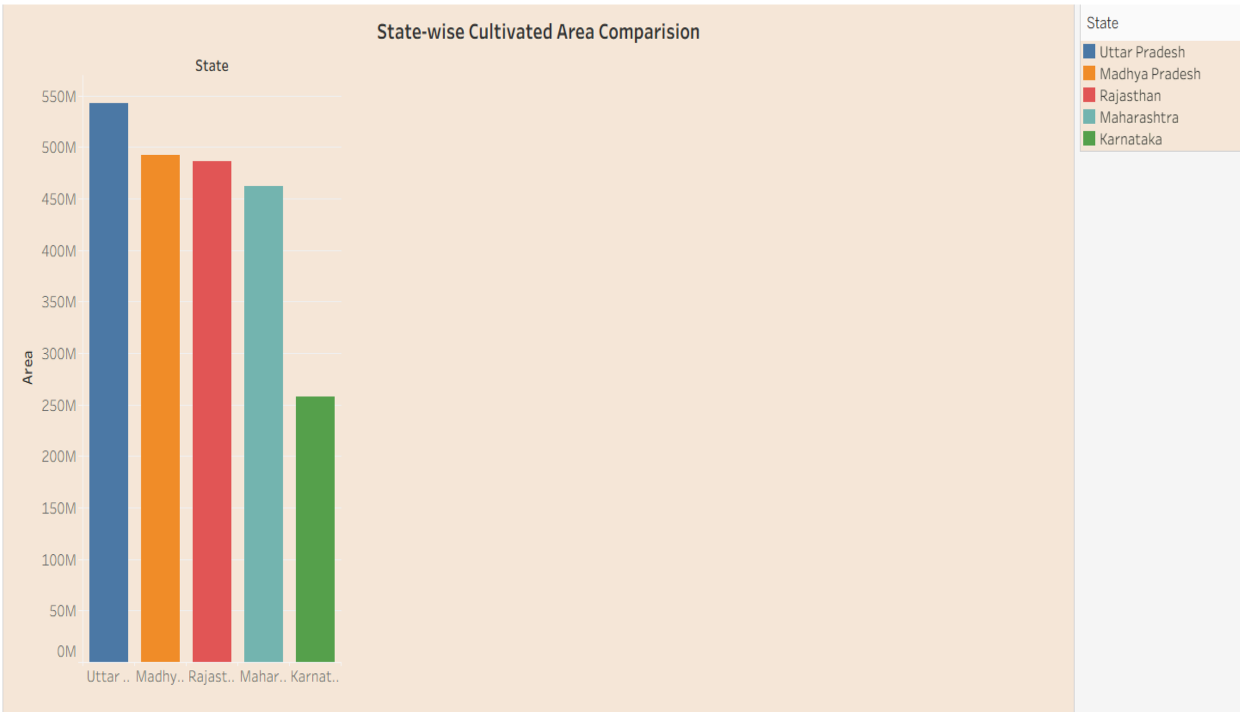
Insight- Rice has the largest cultivated area, followed by wheat, while sugarcane, coconuts, and potato occupy comparatively smaller cultivated areas.



Q.4 Which states have the largest cultivated area in India?

Visualization: Bar chart comparing the cultivated area across different states.

Screenshot of visualization –



Q.5 Which states are the leading contributors to agricultural production in India?

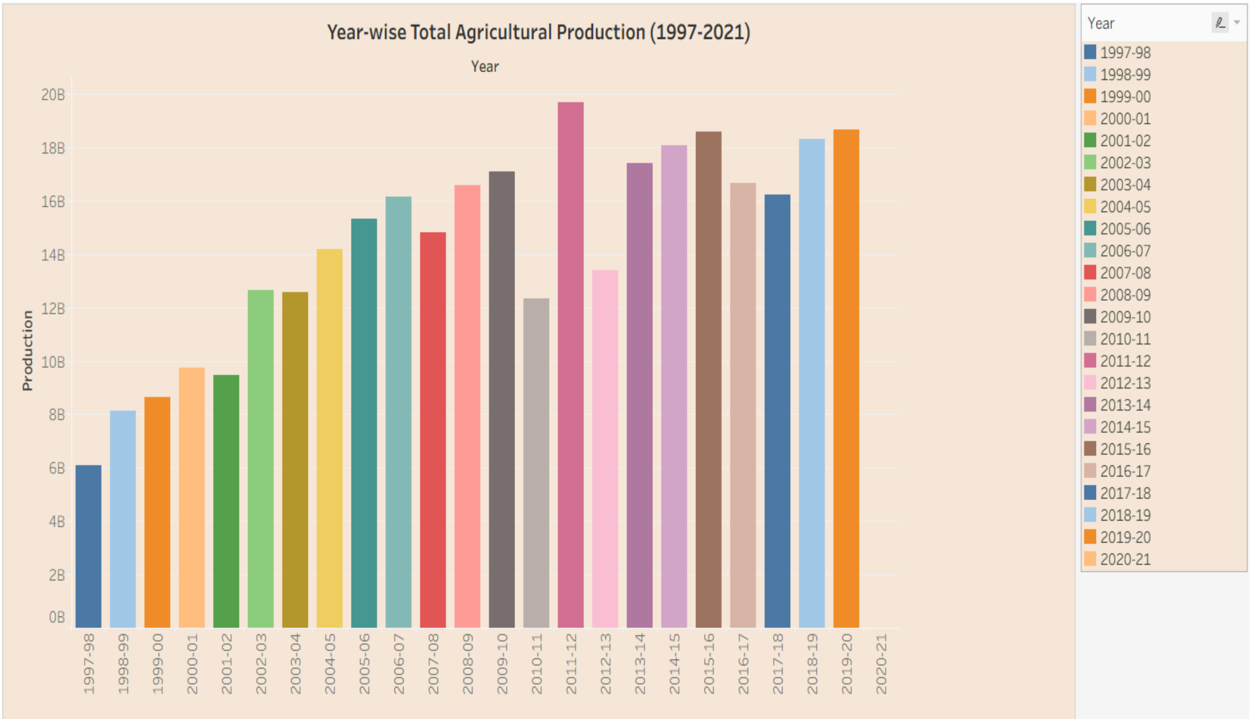
Visualization: Bar chart comparing total agricultural production across selected states.

Screenshot of visualization-

Q.6 How has total agricultural production changed over the years from 1997 to 2021?

Visualization - Bar chart showing year -wise total agricultural production.

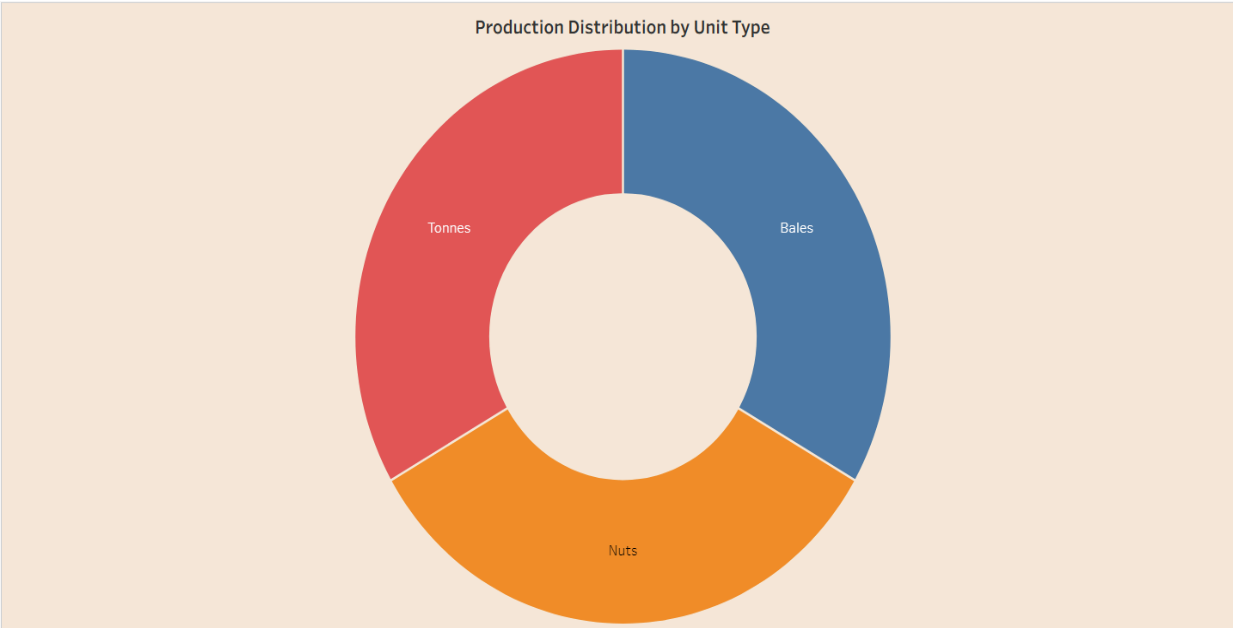
Screenshot of visualization-



Q.7 How is agricultural production distributed across different production unit types?

Visualization- Donut chart showing the distribution of total agricultural production by unit type.

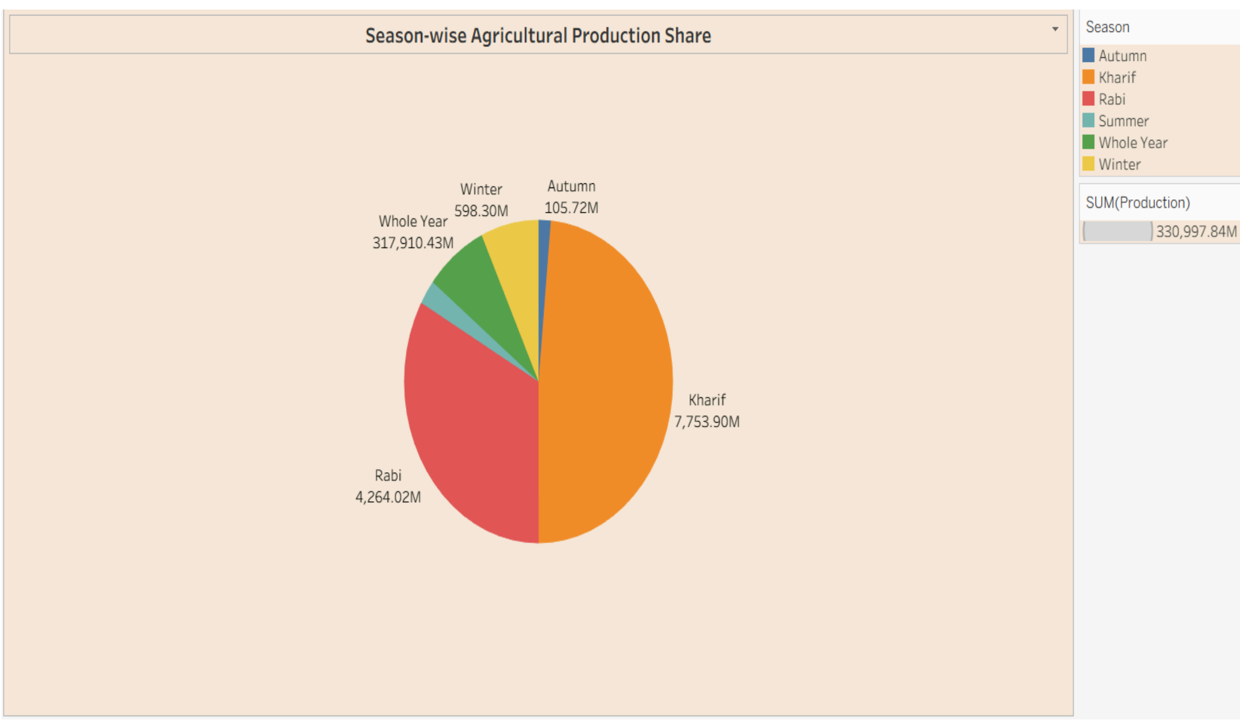
Screenshot of visualization-



Q.8 Which agricultural seasons contribute the most to India's total agricultural production?'

Visualization- Pie chart showing the share of total agricultural production by season.

Screenshot of visualization-



4.2 Developing Visualizations

Different visualization techniques were selected according to the nature of each analytical question.

1. State-wise Agricultural Production Distribution

A geographical map was used to display agricultural production across Indian states.

The map provides a quick visual understanding of the geographical distribution of agricultural production and allows users to identify regions with comparatively higher or lower production.

2. Year-wise Total Agricultural Production

A bar chart was created to compare total agricultural production across different years from 1997 to 2021.

This visualization helps identify changes in production over the study period and makes year-to-year comparisons easier.

3. Trend of Agricultural Yield

A line chart was used to represent the trend of agricultural yield over time.

The line chart makes it easier to identify increases, decreases, fluctuations, and overall patterns in agricultural yield.

4. India Agricultural Performance Summary

A summary section was created to provide important agricultural performance indicators in a compact format.

This allows users to quickly understand the overall scale of production, cultivated area, and yield-related measures without examining every individual chart.

5. Comparison of Cultivated Area by Crop

A horizontal bar chart was used to compare the cultivated area associated with different crops.

This visualization helps identify crops that occupy relatively larger or smaller cultivated areas.

6. State-wise Cultivated Area Comparison

A bar chart was created to compare cultivated areas across selected Indian states.

This helps users understand geographical differences in agricultural land utilization.

7. Production Distribution by Unit Type

A donut chart was used to display the distribution of agricultural production according to production unit types.

The chart provides an easy-to-understand visual comparison of the contribution of different production units.

8. Leading States by Agricultural Production

A bar chart was used to identify and compare the leading states based on agricultural production.

This visualization makes it easier to recognize the states that contribute significantly to India's overall agricultural output.

9. Season-wise Agricultural Production Share

A pie chart was created to show the percentage/share of agricultural production across different agricultural seasons.

This helps users understand the contribution of different seasons such as **Kharif, Rabi, Summer, Autumn, and Whole Year**.

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5. Dashboard

5.1 Dashboard Design File

Dashboard Design

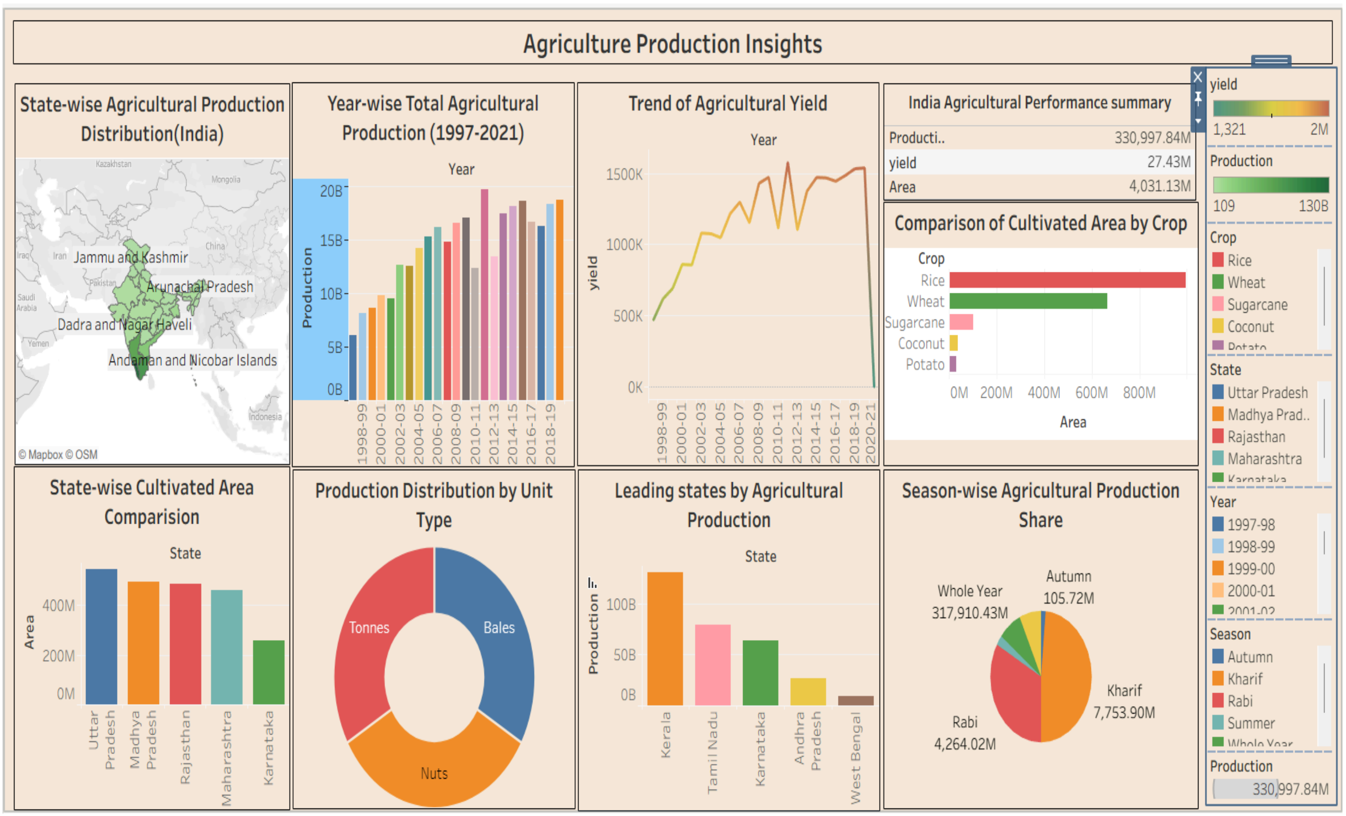
Date	25 August 2026
Team ID	xxxxxx
Project Name	India's agricultural crop production analysis (1997-2021)
Maximum Marks	5 Marks

Activity 1: Interactive and visually appealing dashboards

The dashboard "Agricultural Production Insights" presents an interactive analysis of India's agricultural crop production from 1997-2021.It combines maps, bar charts, line charts, and pie charts to provide insights into production trends cultivated area, stat-wise production, and seasonal production. Filters for year, state, season, and crop allow users to explore the data interactively and compare different aspects of agricultural production.

Checklist-

- Clear and Intuitive lay ought
- Use Appropriate Visualizations
- Colours and Theming
- Interactive filters and slicers
- Responsive Design



Interactive Filters

The dashboard includes interactive filters for:

- **Year**
- **State**
- **Season**
- **Crop**

These filters allow users to select specific categories and dynamically update the dashboard visualizations.

For example, selecting a particular state allows users to analyze its production, cultivated area, seasonal contribution, and other related indicators. Similarly, selecting a particular crop or year allows focused analysis of that category.

Dashboard Design Principles

The following design principles were followed:

- Clear and intuitive layout
- Appropriate selection of chart types
- Consistent visual structure
- Proper use of colours
- Interactive filters and slicers
- Easy comparison between categories
- Responsive dashboard arrangement
- Clear titles and labels
- Logical grouping of related visualizations

The dashboard was designed to provide both an overview of India's agricultural production and the ability to perform detailed interactive analysis.

Activity 2

Publish Dashboard on Tableau Public and Paste the Dashboard Public link below:

https://public.tableau.com/views/Book1_17878090788950/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

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6. Report

6.1 Story Design File

Story

Date	26 August 2026
Team ID	xxxxxx
Project Name	India's agricultural crop production analysis (1997-2021)

Maximum Marks	5 Marks

Observations:

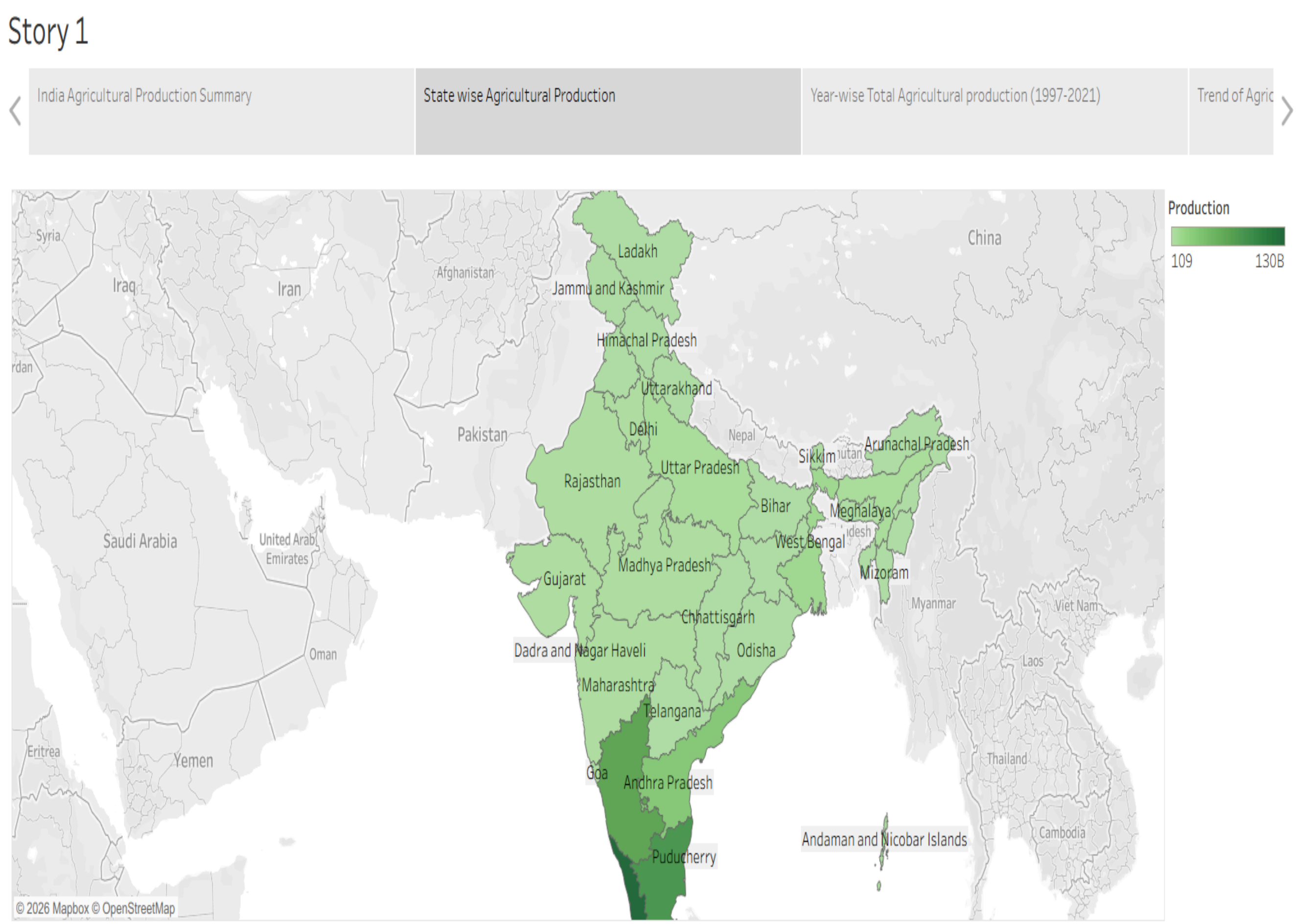
1. **Overall performance**: India's agricultural crop production analysis (1997-2021)

2. **Crop Comparison:** Rice has the largest cultivated area, flowed by wheat.

3. **State comparison**: Uttar Pradesh has the highest cultivated area among the displayed states.

4. **Production Leaders:** Karnataka is the leading state in agricultural production among the states displayed.

5. **Seasonal pattern:** Kharif season contributes the largest share of agricultural, Production followed by Rabi.



A Tableau story was created to present the findings of the project in a structured manner.

The story provides a sequence of analytical views that help users understand agricultural production from different perspectives.

The story focuses on:

1. Overall agricultural production performance.
2. Year-wise production trends.
3. State-wise production comparison.
4. Crop-wise cultivated area.
5. Agricultural yield trends.
6. Leading agricultural-producing states.
7. Season-wise production contribution.
8. Interactive analysis using filters.

The story format makes it easier to communicate the important findings of the dashboard and allows the viewer to understand the analysis step by step.

Activity 2:

Publish Story on Tableau Public and Paste Public link below

https://public.tableau.com/views/Book1_17878090788950/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

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7. Performance Testing

Performance testing was conducted to verify that the dashboard functions correctly and provides interactive analytical capabilities.

7.1 Utilization of Data Filters

The dashboard uses multiple interactive filters.

The main filters used are:

- Year
- State
- Season
- Crop

These filters allow users to narrow down the dataset and analyze specific combinations of agricultural information.

The filters were tested to ensure that the associated charts and dashboard components respond correctly when selections are changed.

The interactive filtering capability improves the usability of the dashboard and allows users to perform customized analysis without modifying the underlying dataset.

7.2 Number of Calculation Fields

Calculated fields were used where required to support analytical measures such as yield-related calculations, aggregated production measures, and other derived indicators.

The calculated fields help transform raw data into meaningful analytical metrics.

The exact number of calculated fields may depend on the final Tableau workbook version submitted with the project.

7.3 Number of Visualizations

The final dashboard contains multiple visualizations covering geographical, temporal, crop-wise, state-wise, seasonal, and production-unit analysis.

The major visualizations include:

1. State-wise production map
2. Year-wise production bar chart
3. Agricultural yield trend line chart
4. Agricultural performance summary
5. Crop-wise cultivated area comparison
6. State-wise cultivated area comparison
7. Production distribution donut chart
8. Leading states bar chart
9. Season-wise production pie chart

Therefore, the dashboard provides a comprehensive combination of charts, map-based visualization, and summary information.

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8. Conclusion / Observation

The **India's Agricultural Crop Production Analysis (1997–2021)** project successfully demonstrates how Tableau can be used to convert a large agricultural dataset into an interactive and understandable visual analytics solution.

The analysis shows that agricultural production varies considerably across years, states, crops, and seasons. The year-wise visualization helps identify the overall changes in production over the 25-year period, while the state-wise map and bar charts provide geographical comparisons.

The cultivated-area analysis shows that different crops occupy different proportions of agricultural land. The yield trend visualization helps in understanding changes in agricultural productivity over time.

The leading-state analysis identifies the major contributors to agricultural production, while the season-wise visualization highlights differences in production contributions among agricultural seasons.

The interactive filters for **Year, State, Season, and Crop** make the dashboard more flexible and useful because users can explore specific parts of the dataset according to their requirements.

Overall, the project demonstrates that data visualization can make complex agricultural information easier to understand and can help users identify important trends, comparisons, and patterns quickly.

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9. Future Scope

The project can be further improved and extended in several ways.

1. **Real-time Data Integration:**
   Future versions can integrate regularly updated agricultural data to provide more current information.
2. **Predictive Analysis:**
   Machine learning techniques can be incorporated to predict future crop production, yield, and cultivated area.
3. **Weather Data Integration:**
   Weather variables such as rainfall, temperature, humidity, and drought conditions can be combined with agricultural data to study their impact on production.
4. **Economic Analysis:**
   Crop prices, market demand, agricultural income, and production costs can be included to study the economic performance of different crops.
5. **Advanced Geographical Analysis:**
   More detailed geographical maps can be developed to analyze district-level agricultural patterns.
6. **Crop Recommendation:**
   Based on historical production, yield, climate, and geographical conditions, a crop recommendation system could be developed.

7. **Forecasting:**
   Time-series forecasting can be used to estimate future production trends for major crops and states.

8. **Advanced Tableau Features:**
   Parameters, set actions, dashboard actions, tooltips, and advanced calculated fields can be used to make the dashboard more interactive.

9. **Automated Reporting:**
   Automated reports can be generated to provide periodic updates on agricultural performance.

10. **Mobile-Friendly Dashboard:**
    The dashboard can be optimized further for mobile devices so that agricultural insights can be accessed conveniently from smartphones and tablets.

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10. Appendix

10.1 Source Code (if any)

This project was primarily developed using **Tableau** for data analysis and visualization. Therefore, traditional programming source code was not the primary component of the project.

The Tableau workbook contains the data connection, worksheets, calculated fields, filters, dashboards, and story used for the analysis.

If any preprocessing or supporting calculations were performed outside Tableau, the corresponding files or scripts can be included in the project repository.

10.2 GitHub & Project Demo Link

GitHub Repository

The complete project files, Tableau workbook, documentation, and supporting project materials can be made available through the project's GitHub repository.

GitHub Link:
https://github.com/rajattiwari2301/India-s-agricultural-crop-production-analysis-1997-2021-

Tableau Public / Project Link

The final interactive Tableau dashboard can be accessed through the following link:

Tableau Public Link:
https://public.tableau.com/views/SalesDashboard_17247660582000/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Project Demo Video

A demonstration video explaining the dashboard, visualizations, filters, insights, and project workflow can be submitted along with the project.

Demo Video Link:
https://drive.google.com/file/d/1wfAMQHvMd02UAV2qPk3Lqzu9uRtVBUkE/view?usp=drivesdk

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Final Project Summary

The project **"India's Agricultural Crop Production Analysis (1997–2021)"** provides an interactive visual analysis of historical agricultural production data in India. Using Tableau, the project transforms raw data into maps, bar charts, line charts, pie charts, donut charts, and summary indicators.

The dashboard enables users to analyze agricultural production according to **year, state, crop, season, cultivated area, production unit, and yield**. Interactive filters make it possible to perform customized analysis and compare different agricultural dimensions.

The project demonstrates the practical application of **data analytics, data visualization, dashboard design, and interactive reporting** to an important real-world domain. The insights obtained from the analysis can support better understanding of India's agricultural production patterns and can provide a foundation for future predictive and decision-support applications.